

Lecture 4: Calculus and Linear Algebra

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Course content (math tapas)

Sequences
and series

Functions
of one variable

Differentiation

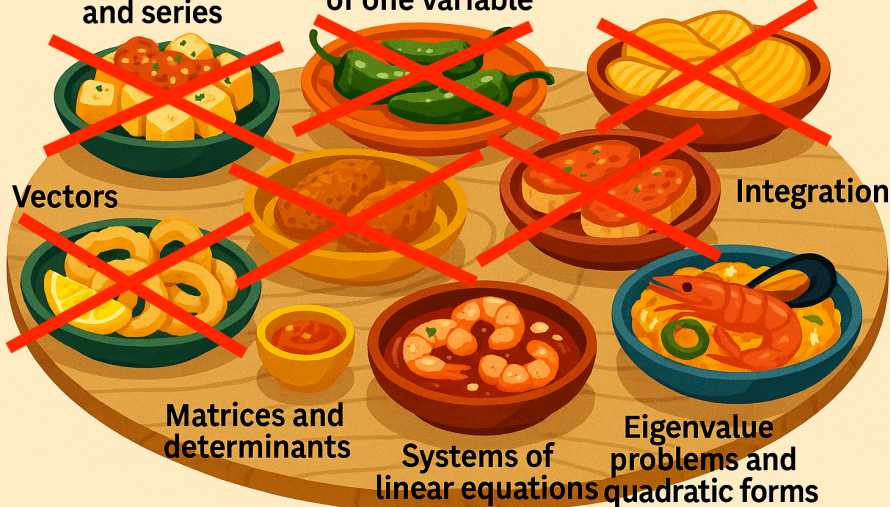
Vectors

Integration

Matrices and
determinants

Systems of
linear equations

Eigenvalue
problems and
quadratic forms



Linear subspaces

Linear subspace

The set $V \subset \mathbb{R}^n$ is a **linear subspace** of $\mathbb{R}^n$ if

$$\forall \mathbf{u}, \mathbf{v} \in V \quad \forall a, b \in \mathbb{R} \quad a\mathbf{u} + b\mathbf{v} \in V.$$

(closure under taking linear combinations)

There are two ways we can describe a linear subspace V :

- $V = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ for some vectors $\mathbf{v}_1, \dots, \mathbf{v}_m \in V$.
- Solution set of $A\mathbf{x} = \mathbf{0}_m$ for some $A \in \mathbb{R}^{m \times n}$.

Some examples:

- kernel of $A \in \mathbb{R}^{m \times n}$
- image of A
- Points in $\mathbb{R}^3$ satisfying $x_1 + x_2 + x_3 = 0$.

Why subspaces must have this form?

Let $V \subseteq \mathbb{R}^n$ be a linear subspace.

Take any nonzero $\mathbf{v}_1 \in V$. Then $\text{span}(\mathbf{v}_1) \subseteq V$.

If $\text{span}(\mathbf{v}_1) \subseteq V$, we are done.

If not, take any nonzero $\mathbf{v}_2 \in V \setminus \text{span}(\mathbf{v}_1)$.

Then $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent and again $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\} \subseteq V$.

We proceed iteratively until we reach step m such that

$$V = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\}.$$

This procedure must stop as we cannot have more than n linearly independent vectors in $V \subseteq \mathbb{R}^n$.

System of linear equations

Systems of linear equations

Let A be an $m \times n$ matrix, $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ unknown, and $\mathbf{b} \in \mathbb{R}^m$ given. The system

$$A\mathbf{x} = \mathbf{b}$$

has the **vector form**

$$\sum_{j=1}^n x_j \mathbf{a}_j = \mathbf{b},$$

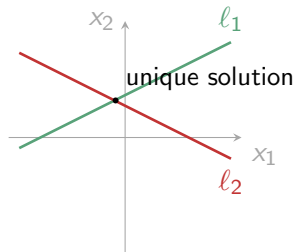
where $\mathbf{a}_1, \dots, \mathbf{a}_n$ are the columns of A .

If $\mathbf{b} = \mathbf{0}$ the system is **homogeneous**; otherwise it is **inhomogeneous**.

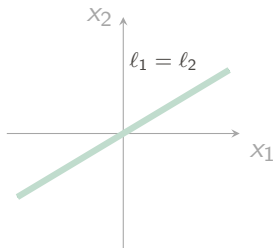
An inhomogeneous system can be **inconsistent** (no solution) or **consistent** (at least one solution).

A homogeneous system always has $\mathbf{x} = \mathbf{0}$ as a solution.

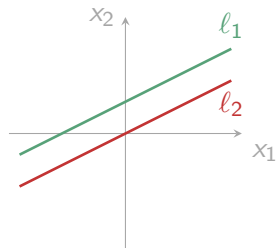
Geometry of 2×2 linear systems



Unique solution



Infinitely many solutions



No solution (inconsistent)

Key: Lines are the solution sets of single linear equations. Intersecting $\Rightarrow$ one point.
Coincident $\Rightarrow$ all points. Parallel $\Rightarrow$ none.

Augmented matrix; consistency

The **augmented matrix** of the linear system $A\mathbf{x} = \mathbf{b}$ is

$$A_{\mathbf{b}} = [A \ \mathbf{b}] \in \mathbb{R}^{m \times (n+1)}.$$

Remark: Either $\text{rank}(A_{\mathbf{b}}) = \text{rank}(A)$ or $\text{rank}(A_{\mathbf{b}}) = \text{rank}(A) + 1$.

Theorem (Rouché–Capelli)

$A\mathbf{x} = \mathbf{b}$ is **consistent** $\iff \text{rank}(A_{\mathbf{b}}) = \text{rank}(A)$ (equivalently $\mathbf{b} \in \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$).

Moreover:

1. If $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) = n$, the solution is **unique**. **Q:** What is it?
2. If $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) < n$, there are **infinitely many** solutions.

The rank tells us how many variables are **pivots** (determined by equations).

The remaining variables can be chosen freely, hence called **free variables**.

Solving with Gaussian elimination

Row echelon forms

A matrix is in **row echelon form (REF)** if:

1. All nonzero rows appear above any zero rows.
2. Each **pivot** (leading entry in a row) is to the right of the pivot in the row above it.
3. All entries below a pivot are zero.

It is in **reduced row echelon form (RREF)** if, in addition:

1. Each pivot is equal to 1.
2. Each pivot is the only nonzero entry in its column.

Example:

$$\begin{pmatrix} \mathbf{1} & 2 & 0 & 3 \\ 0 & \mathbf{1} & 4 & -1 \\ 0 & 0 & 0 & \mathbf{1} \end{pmatrix} \quad (\text{REF, not RREF}). \qquad \begin{pmatrix} \mathbf{1} & 0 & 0 & 2 \\ 0 & \mathbf{1} & 0 & -3 \\ 0 & 0 & \mathbf{1} & 5 \end{pmatrix} \quad (\text{RREF}).$$

Gaussian elimination and RREF

Theorem

By applying elementary row operations to the augmented matrix $A_{\mathbf{b}}$, any system $A\mathbf{x} = \mathbf{b}$ can be transformed into an equivalent system in **reduced row echelon form (RREF)**.

$$\begin{array}{rcl} x_1 + x_2 + x_3 & = & 3 \\ x_1 - x_2 + 2x_3 & = & 2 \\ 4x_1 + 6x_2 - x_3 & = & 9 \end{array} \iff \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right)$$

So the unique solution is $\mathbf{x} = (1, 1, 1)$.

Why equivalence holds:

$A\mathbf{x} = \mathbf{b} \iff A_{\mathbf{b}} \begin{bmatrix} \mathbf{x} \\ -1 \end{bmatrix} = \mathbf{0}$. Row operations correspond to multiplying $A_{\mathbf{b}}$ on the left by an invertible matrix C , so the solution set is unchanged.

A homogeneous example

Consider

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 0,$$

$$2x_1 + 2x_2 + 2x_3 + 2x_4 - x_5 - x_6 = 0,$$

$$3x_1 + 3x_2 - x_3 - x_4 - x_5 - x_6 = 0.$$

Row-reducing the coefficient matrix gives

$$\begin{pmatrix} \color{red}{1} & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \color{red}{1} & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \color{red}{1} & 1 \end{pmatrix}.$$

So there are 3 pivots, $\text{rank}(A) = 3$, and the solution space has dimension $6 - \text{rank}(A) = 3$:

$$(x_1, -x_1, x_3, -x_3, x_5, -x_5).$$

A basis is $\{(1, -1, 0, 0, 0, 0), (0, 0, 1, -1, 0, 0), (0, 0, 0, 0, 1, -1)\}$.

General solution structure

Theorem

Every solution of $A\mathbf{x} = \mathbf{b}$ can be written as $\mathbf{x} = \mathbf{x}_h + \mathbf{x}_p$, where $\mathbf{x}_h$ is the general solution of the homogeneous system $A\mathbf{x} = \mathbf{0}$ and $\mathbf{x}_p$ is any particular solution of $A\mathbf{x} = \mathbf{b}$.

Corollary: If $A\mathbf{x} = \mathbf{b}$ is consistent, then the set of solutions is an affine subspace parallel to $\ker(A)$.

Example: Three identical equations

$$x_1 + x_2 + x_3 = 1$$

reduce to

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \rightsquigarrow \begin{array}{l} x_1 = 1 - x_2 - x_3, \\ x_2, x_3 \text{ free.} \end{array}$$

So

$$\mathbf{x} = (1, 0, 0) + \lambda_1(-1, 1, 0) + \lambda_2(-1, 0, 1).$$

Recipe: solving $A\mathbf{x} = \mathbf{b}$ with Gaussian elimination

1. Write down the augmented matrix $[A \mid \mathbf{b}]$.
2. Use row operations to reduce to RREF.
3. Identify pivot and free variables.
4. Express the solution: $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$.

Always

- The number of pivot variables is the rank of A ,
- the number of free variables $= n - \text{rank}(A)$,
- the free variables parametrize the solution set.

Two quick examples

1. For $A = \begin{pmatrix} 1 & 2 & 0 \\ 4 & 6 & 2 \\ 3 & 2 & 4 \end{pmatrix}$, RREF gives $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$. So $\text{rank}(A) = 2$, $\dim \ker(A) = 1$, and

$$\ker(A) = \{\lambda(-2, 1, 1) : \lambda \in \mathbb{R}\}.$$

2. For $\begin{cases} x_1 + x_2 + x_3 = -1 \\ x_1 + 2x_2 + 4x_3 = 2 \end{cases}$, the augmented matrix reduces to $\left(\begin{array}{ccc|c} 1 & 0 & -2 & -4 \\ 0 & 1 & 3 & 3 \end{array} \right)$. So $\text{rank}(A) = 2$, one free variable $x_3 = \lambda$, and

$$\mathbf{x} = (-4, 3, 0) + \lambda(2, -3, 1).$$

Finding a basis for a span

Let

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 0 \\ -1 \\ -2 \end{pmatrix}.$$

Put them as columns of a matrix:

$$M = \begin{pmatrix} 2 & -1 & 0 \\ 3 & 0 & -1 \\ 0 & 3 & -2 \end{pmatrix}.$$

Row-reduction gives $M \sim \begin{pmatrix} 1 & 0 & -1/3 \\ 0 & 1 & -2/3 \\ 0 & 0 & 0 \end{pmatrix}.$

Thus $\text{rank}(M) = 2$ and there is one linear dependence: $3\mathbf{v}_1 - 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$.
A basis for $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ can be taken as the first two pivot columns, e.g.

$$\{\mathbf{v}_1, \mathbf{v}_2\}.$$

Quick recap

- **Homogeneous system:** $Ax = 0$. Always has at least the trivial solution.
- **Inhomogeneous system:** $Ax = b$. May or may not be solvable.
- **Augmented matrix:** $[A \mid b]$, i.e. stick b as a new column.
- **Rank:** Number of pivot columns after row reduction = dimension of the column space.

Standing on the shoulders of giants



“If I have seen further, it is by standing on the shoulders of giants.” **Isaac Newton (1675).**

Some of the giants of linear/matrix algebra:

- Gauss (Gaussian elimination, least squares, determinants)
- Cauchy (Cauchy–Schwarz, eigenvalues, determinants)
- Laplace (Laplace expansion of determinants)
- Cayley (Cayley–Hamilton theorem, matrix theory)
- Sylvester (matrix rank, invariant theory)
- Jordan (Jordan canonical form)
- Frobenius (matrix factorizations, linear algebra foundations)

(pictured here as the base of a *castell*)

Chapter 8: Eigenvalues and Quadratic Forms

Eigenvalues/eigenvectors appear in growth models, Markov chains, PCA, and stability of dynamical systems.

Quadratic forms drive optimization via second-order conditions.

Read Chapter 10 of Werner–Sotskov (optional: Simon–Blume, Ch. 7).

Exercises 10.2; 10.5 A,B,C; 10.6(a,b).

Eigenvalues and eigenvectors

Eigenvalues and eigenvectors: definitions

For $A \in \mathbb{R}^{n \times n}$, a scalar λ is an **eigenvalue** if there exists $\mathbf{x} \neq \mathbf{0}$ with

$$A\mathbf{x} = \lambda\mathbf{x} \iff (A - \lambda I_n)\mathbf{x} = \mathbf{0}.$$

Then $\mathbf{x}$ is an **eigenvector** for λ .

Characterization:

$$\lambda \text{ eigenvalue} \iff \text{rank}(A - \lambda I_n) < n \iff \det(A - \lambda I_n) = 0.$$

The **characteristic polynomial** $P(\lambda) = \det(A - \lambda I_n)$ has degree n ; its real roots (at most n) are the real eigenvalues.

Procedure:

1. Solve $\det(A - \lambda I_n) = 0$ for eigenvalues.
2. For each λ , solve $(A - \lambda I_n)\mathbf{x} = \mathbf{0}$ for eigenvectors.

e.g. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$, $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

Why eigenvalues matter (at a glance)

If $A\mathbf{x} = \lambda\mathbf{x}$ then $A^k\mathbf{x} = A^{k-1}(A\mathbf{x}) = \lambda A^{k-1}\mathbf{x} = \dots = \lambda^k\mathbf{x}$.

- **Markov chains:** steady state = eigenvector for $\lambda = 1$.
- **Growth/dynamics:** long-run rate $\approx$ spectral radius $\rho(A)$.
- **PCA:** principal directions = eigenvectors of covariance; explained variances = eigenvalues.
- **Quadratic forms:** curvature of $\mathbf{x}^\top A\mathbf{x}$ aligned with eigenvectors; definiteness from eigenvalue signs.

Eigenvalues and eigenvectors: migration example

Two regions: city (x_t^C) and countryside (x_t^R). Let $\mathbf{x}_t = (x_t^C, x_t^R)^\top$. Assume next-year proportions follow a **column-stochastic** transition:

$$\mathbf{x}_{t+1} = A \mathbf{x}_t, \quad A = \begin{pmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{pmatrix},$$

so each column sums to 1 (fractions of people who *arrive* from each region).

A **stable composition** $\mathbf{u}$ satisfies $A\mathbf{u} = \mathbf{u}$ and $u_1 + u_2 = 1$:

$$(A - I)\mathbf{u} = 0 \Rightarrow \mathbf{u} = \begin{pmatrix} \frac{2}{3} \\ \frac{1}{3} \end{pmatrix}.$$

Spectrum: $\lambda_1 = 1$, $\lambda_2 = 0.4$.

Any initial $\mathbf{x}_0$ with positive entries converges to $\mathbf{u}$.

- Check $\mathbf{v} = (1, -1)$ is the 0.4-eigenvector.
- $\mathbf{x}_0$ can be written as $\mathbf{x}_0 = \mathbf{u} + \lambda \mathbf{v}$
- $\mathbf{x}_t = A^t(\mathbf{u} + \lambda \mathbf{v}) = \mathbf{u} + \lambda \cdot 0.4^t \cdot \mathbf{v}$.
- $\|\mathbf{x}_t - \mathbf{u}\| = |\lambda| \cdot 0.4^t \sqrt{2} \rightarrow 0$.

Basic spectral facts

For $A \in \mathbb{R}^{n \times n}$

1. Eigenvectors associated with distinct eigenvalues are **linearly independent**.
2. If A is **symmetric**, then all eigenvalues are real and eigenvectors for distinct eigenvalues are **orthogonal** (Spectral Theorem).

Example:

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

Then $P(\lambda) = \lambda^2 - 2\lambda - 3$, so $\lambda_1 = 3$, $\lambda_2 = -1$ with eigenvectors along $(1, 1)$ and $(-1, 1)$, respectively.

Matrix diagonalization

Diagonalizable or not?

Definitions:

- $A \in \mathbb{R}^{n \times n}$ is **diagonalizable** if there exists an invertible P such that

$$P^{-1}AP = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Equivalently, A has a basis of eigenvectors.

- For each eigenvalue λ , the **eigenspace** is

$$E_\lambda = \ker(A - \lambda I).$$

Its dimension is the **geometric multiplicity**.

- The number of times λ appears as a root of the characteristic polynomial is its **algebraic multiplicity**.

Why it matters: diagonalization simplifies A^k , e^A .

Fact:

A is diagonalizable $\iff \sum_{\lambda} \dim E_{\lambda} = n$.

Example 1 (diagonalizable).

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Characteristic polynomial: $(2 - \lambda)^2 - 1$. Distinct eigenvalues $\Rightarrow$ eigenvectors span $\mathbb{R}^2$. So A is diagonalizable: $A = PDP^{-1}$ with $D = \text{diag}(3, 1)$ and

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Example 2 (defective).

$$B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Characteristic polynomial: $(\lambda - 1)^2$ (algebraic multiplicity 2). Eigenspace: $\ker(B - I) = \text{span}\{(1, 0)\}$ has dimension 1. So B is **not** diagonalizable.

Repeated eigenvalues

Consider

$$A = \begin{pmatrix} 0 & -1 & 1 \\ -7 & 0 & 5 \\ -5 & -2 & 5 \end{pmatrix}.$$

The characteristic polynomial is

$$p_A(\lambda) = \lambda^3 - 5\lambda^2 + 8\lambda - 4 = (\lambda - 2)^2(\lambda - 1).$$

Hence the eigenvalues are $\lambda = 2$ (algebraic mult. 2) and $\lambda = 1$ (algebraic mult. 1).

Eigenspaces:

$$E_2 = \ker(A - 2I) = \text{span}\{(1, -1, 1)\}, \quad E_1 = \ker(A - I) = \text{span}\{(2, 1, 3)\}.$$

Since $\dim E_2 = 1 < 2$, A is defective.

Spectral decomposition and applications

Spectral decomposition (symmetric case)

Theorem (Spectral Theorem)

If $A \in \mathbb{R}^{n \times n}$ is symmetric, then there exist an orthogonal $U \in \mathbb{R}^{n \times n}$ and a diagonal $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ such that

$$A = U\Lambda U^\top, \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n.$$

The columns of U are orthonormal eigenvectors of A .

Application: Principal Component Analysis (PCA)

We observe n data points $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$.

Step 1 (Centering): Arrange the data in a matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ with rows $\mathbf{x}_1, \dots, \mathbf{x}_n$. Define the centering matrix $H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ and then the **centered data** is $H\mathbf{X}$.
(its rows are the centered data points $\mathbf{x}_i - \bar{\mathbf{x}}$)

Step 2 (Sample covariance):

$$S = \frac{1}{n}(H\mathbf{X})^\top(H\mathbf{X}).$$

Step 3 (Spectral decomposition):

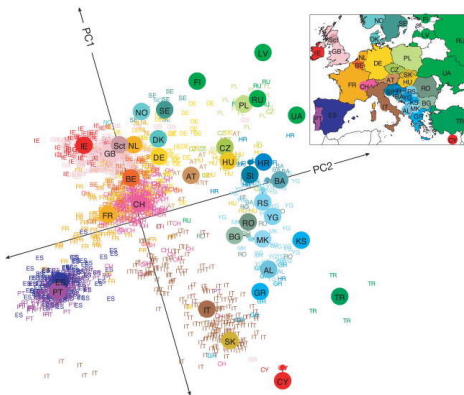
$$S = U\Lambda U^\top \quad \text{and define } \mathbf{Y} = H\mathbf{X}U \quad (\text{rotate the centered data}).$$

The sample covariance of $\mathbf{Y}$ is $\frac{1}{n}\mathbf{Y}^\top\mathbf{Y} = U^\top S U = \Lambda$ (diagonal! and so $\mathbf{y}_i$ are uncorrelated).

The columns of U are **principal directions** and diagonal entries of Λ are the **explained variances**.

Sorting eigenvalues in decreasing order ranks components by variance captured.

PCA: why it matters



Dimensionality reduction: compress high-dimensional data into a few directions that preserve most variability.

Noise filtering: small eigenvalues $\rightarrow$ directions with little signal.

Visualization: projecting onto first 2 PCs often reveals hidden structure.

Nice example: genetic variation in Europeans — plotting individuals on the first two PCs of SNP data *recovers the geographic map of Europe!*

Numerical toy example

Covariance matrix:

$$S = \begin{pmatrix} 72.9 & 87.5 & 6.875 \\ 87.5 & 108.3 & 8.75 \\ 6.875 & 8.75 & 0.73 \end{pmatrix}.$$

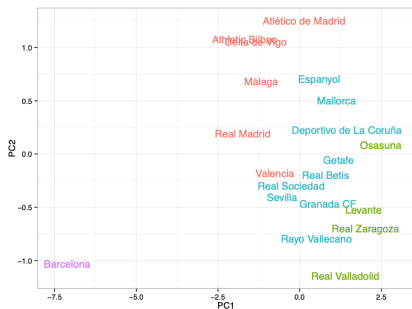
Spectral decomposition:

$$S = U\Lambda U^{\top}, \quad \Lambda = \text{diag}(180, 1.38, 0.005).$$

⇒ The first component explains > 99% of the variance; the last can be safely ignored.

PCA & Unique football styles

“Mille viae ducunt Barcinonem” (A thousand roads lead to Barcelona)



“Searching for a Unique Style in Soccer” (Gyarmati, Kwak & Rodriguez, 2014), studies sequential pass patterns in a team’s passing.

They defined “flow motifs” as statically significant pass subsequences (e.g., $A \rightarrow B \rightarrow C \rightarrow A$).

Applied motif analysis to compare styles across teams. The data are projected to the first two principal components.

Quadratic forms

Quadratic forms

A quadratic form is

$$Q(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x}, \quad A \in \mathbb{R}^{n \times n}.$$

Why symmetric? If $a_{ij} \neq a_{ji}$, the asymmetric part cancels:

$$\mathbf{x}^\top A \mathbf{x} = \mathbf{x}^\top \frac{1}{2}(A + A^\top) \mathbf{x}.$$

So every quadratic form has a unique symmetric representative.

Examples:

$$Q(x, y) = x^2 + 5xy + 4y^2$$

comes from $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ or equivalently $\begin{pmatrix} 1 & 2.5 \\ 2.5 & 4 \end{pmatrix}$.

Geometry: level sets $\{\mathbf{x} : Q(\mathbf{x}) = c\}$ are ellipses, hyperbolas, or paraboloids.

Signs of quadratic forms

For symmetric A :

- **Positive definite (PD)**: $x^\top Ax > 0$ for all $x \neq 0$.
- **Negative definite (ND)**: $x^\top Ax < 0$ for all $x \neq 0$.
- **Semidefinite**: always ≥ 0 (PSD) or ≤ 0 (NSD).
- **Indefinite**: can take both signs.

Why it matters:

- In optimization, a PD Hessian $\Rightarrow$ strict local minimum.
- In economics, concave utility functions $\Rightarrow$ risk aversion.

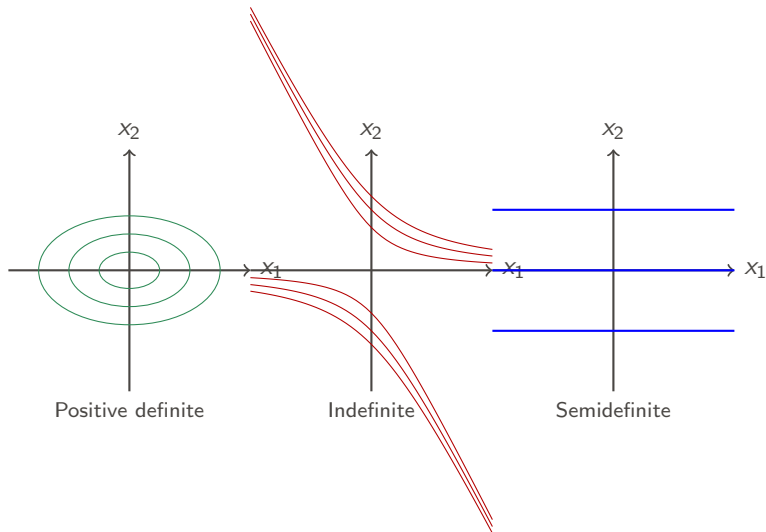
Example:

$$x^\top \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} x = (x - y)^2 \geq 0$$

so the matrix is PSD.

Geometry of quadratic forms

Level sets $\{x : Q(x) = c\}$ illustrate definiteness:



Ellipses $\rightarrow$ PD, Hyperbolas $\rightarrow$ Indefinite, Parallel lines $\rightarrow$ Semidefinite.

Eigenvalue and minor tests

Spectral test (symmetric A):

- PD $\iff$ all eigenvalues > 0 ; ND $\iff$ all < 0 ;
- PSD $\iff$ all ≥ 0 ; NSD $\iff$ all ≤ 0 ;
- Indefinite $\iff$ eigenvalues of different signs.

Sylvester's criterion (leading principal minors D_k):

- PD $\iff D_k > 0$ for $k = 1, \dots, n$;
- ND $\iff (-1)^k D_k > 0$ for $k = 1, \dots, n$.

Example: $A = \begin{pmatrix} -3 & 2 & 0 \\ 2 & -3 & 0 \\ 0 & 0 & -5 \end{pmatrix}$ has $D_1 = -3$, $D_2 = 5$, $D_3 = -25 \Rightarrow$
negative definite.

Examples of quadratic forms

1. $A = \begin{pmatrix} 1 & 4 & 6 \\ 4 & 2 & 1 \\ 6 & 1 & 6 \end{pmatrix} \Rightarrow$ indefinite (mixed signs of minors).

2. $A = \begin{pmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{pmatrix} \Rightarrow$ positive definite.

3. $Q(x, y, z) = xy + yz \Rightarrow$ indefinite (matrix has both positive and negative eigenvalues).

Takeaway: sign of eigenvalues = curvature type.

Chapter 9: Functions of several variables

Many economic and data science models depend on **several variables simultaneously**.

Why this matters: In economics, technologies and preferences are multivariate (production, utility). In data science/ML, high-dimensional loss functions $L(\theta_1, \dots, \theta_d)$ are optimized using gradients/Hessians.

Examples:

- **Economics:** Cobb–Douglas production $Y = K^\alpha L^{1-\alpha}$, utility $U(x, y)$.
- **Data science:** Loss functions $L(\theta)$ with $d \gg 1$ parameters.

Reading: Werner–Sotskov (Ch. 11); Simon–Blume (Chs. 14, 17).

Exercises: 11.11(a), 11.21, 11.22 (Werner–Sotskov).

What is a multivariable function?

A function of n variables is a map

$$f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}, \quad \mathbf{x} = (x_1, \dots, x_n) \mapsto f(\mathbf{x}),$$

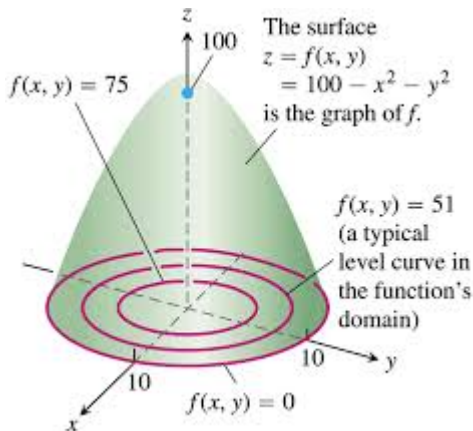
typically with D open. When discussing gradients/tangent planes we assume $f \in C^1$ on D .

- For $n = 2$: the graph $z = f(x, y)$ is a surface in $\mathbb{R}^3$.
- Level curves (contours): $\{(x, y) \in D : f(x, y) = c\}$.
- Sub/superlevel sets: $\{f \leq c\}, \{f \geq c\}$ — useful for (quasi-)convexity.
- For $n > 2$: use slices or projections to visualize.

Example: quadratic function

$$f(x, y) = 100 - x^2 - y^2$$

- Graph of f : paraboloid.
- Level curves: concentric circles.



Economic example: Cobb–Douglas

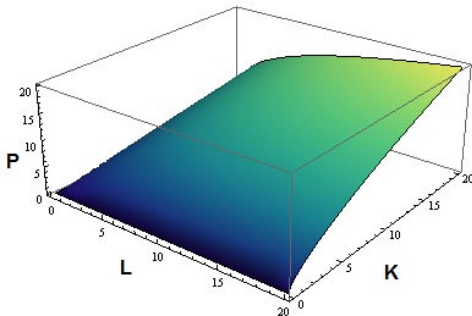
Cobb–Douglas production function

$$P(L, K) = b L^{\alpha} K^{\beta}.$$

P = output, L = labour, K = capital, b = total factor productivity, α, β = output elasticities.

Domain:

$$D = \{(L, K) \in \mathbb{R}^2 : L \geq 0, K \geq 0\}.$$



Returns to scale in Cobb–Douglas

For $t > 0$:

$$P(tL, tK) = b(tL)^\alpha (tK)^\beta = t^{\alpha+\beta} P(L, K).$$

- $\alpha + \beta = 1$: constant returns to scale,
- $\alpha + \beta < 1$: decreasing returns,
- $\alpha + \beta > 1$: increasing returns.

Multivariate Gaussian density

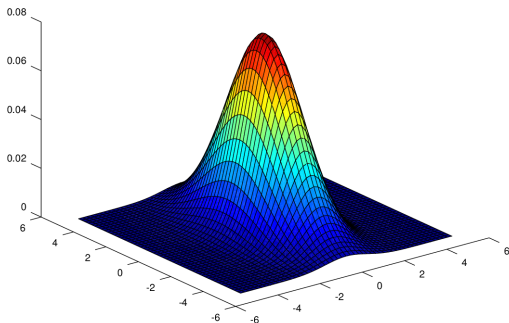
A random vector $\mathbf{X} \in \mathbb{R}^d$ is **multivariate normal** with mean $\mathbf{m}$ and $\Sigma \succ 0$ if

$$f(\mathbf{x}) = (2\pi)^{-d/2} (\det \Sigma)^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{m})^\top \Sigma^{-1}(\mathbf{x} - \mathbf{m})\right).$$

- $\mathbf{m} = E(\mathbf{X})$, $\Sigma = \text{Cov}(\mathbf{X}) \succ 0$.
- Depends on *Mahalanobis* distance
$$\|\mathbf{x} - \mathbf{m}\|_\Sigma = \sqrt{(\mathbf{x} - \mathbf{m})^\top \Sigma^{-1}(\mathbf{x} - \mathbf{m})}.$$
- Level sets are ellipsoids $(\mathbf{x} - \mathbf{m})^\top \Sigma^{-1}(\mathbf{x} - \mathbf{m}) = \text{const.}$

Multivariate Gaussian density

Example: $d = 2$, $\mathbf{m} = \mathbf{0}$, variances $\sigma_1^2 = 1$, $\sigma_2^2 = 4$ (zero covariance).



Partial derivatives

Definition

For $f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$, the partial derivatives at (x, y) are

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}, \quad f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y + h) - f(x, y)}{h}$$

when these limits exist. (We also use more standard notation $\frac{\partial f}{\partial x}(x, y)$, $\frac{\partial f}{\partial y}(x, y)$)

Equivalently, fix y and define $g(x) = f(x, y)$. Then $f_x(x, y) = g'(x)$.

Example (marginal costs): if

$$C(x, y) = 200 + 22x + 16y^{3/2},$$

then $C_x(x, y) = 22$ and $C_y(x, y) = 24\sqrt{y}$.

Cobb–Douglas: marginal productivities

For $P(L, K) = b L^\alpha K^\beta$,

$$\frac{\partial P}{\partial L} = \alpha \frac{P}{L}, \quad \frac{\partial P}{\partial K} = \beta \frac{P}{K}.$$

Elasticities: $\varepsilon_{P,L} = \frac{\partial P}{\partial L} \cdot \frac{L}{P} = \alpha$, $\varepsilon_{P,K} = \beta$.

Tangent plane and linear approximation

Geometrically, $f_x(x_0, y_0)$ is the slope of the tangent to the curve cut by $y = y_0$ at (x_0, y_0, z_0) , $z_0 = f(x_0, y_0)$.

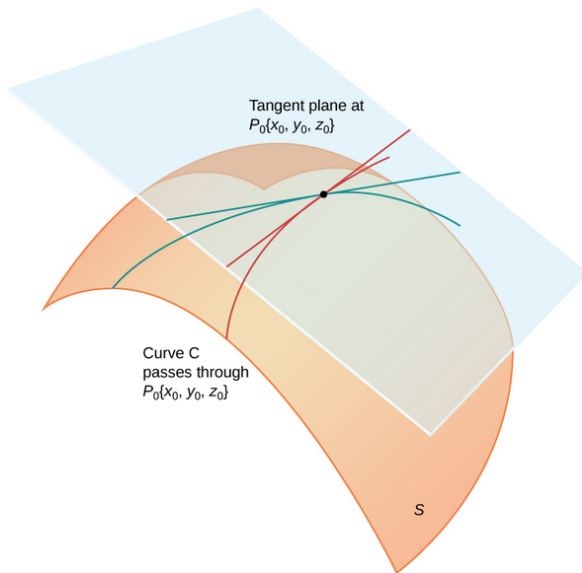
Tangent plane at (x_0, y_0, z_0) :

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

If $f \in C^1$, then for $\Delta = (\Delta x, \Delta y)$,

$$f(x_0 + \Delta x, y_0 + \Delta y) = f(x_0, y_0) + f_x \Delta x + f_y \Delta y + o(\|\Delta\|).$$

Tangent plane (visual)



Higher partial derivatives

Higher partials are iterated derivatives, e.g.

$$f_{xy}(x, y) = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x}(x, y) \right).$$

Young's theorem: If f_{xy} and f_{yx} are continuous near a point, then $f_{xy} = f_{yx}$ there.

Example: $f(x, y) = x^2y + e^{xy}$ gives

$$f_{xy} = 2x + e^{xy}(1 + xy) = f_{yx}.$$

The gradient and linear approximation

For $\mathbf{x} \in \mathbb{R}^n$, the **gradient** is

$$\nabla f(\mathbf{x}) = (f_{x_1}(\mathbf{x}), \dots, f_{x_n}(\mathbf{x})).$$

Best linear approximation

If f is a C^1 -function, then for $\mathbf{h} \in \mathbb{R}^n$,

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{h} + o(\|\mathbf{h}\|).$$

So the gradient gives the **best linear approximation** of f near $\mathbf{x}$.

Geometry: ∇f is normal to level sets $f(\mathbf{x}) = c$

Why is the gradient the direction of steepest ascent?

Take $\mathbf{h} = t\mathbf{u}$ with $\|\mathbf{u}\| = 1$, $t > 0$ small. Then

$$f(\mathbf{x} + t\mathbf{u}) = f(\mathbf{x}) + t \langle \nabla f(\mathbf{x}), \mathbf{u} \rangle + o(t).$$

The instantaneous rate of change in direction $\mathbf{u}$ is

$$D_{\mathbf{u}}f(\mathbf{x}) := \lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{u}) - f(\mathbf{x})}{t} = \langle \nabla f(\mathbf{x}), \mathbf{u} \rangle.$$

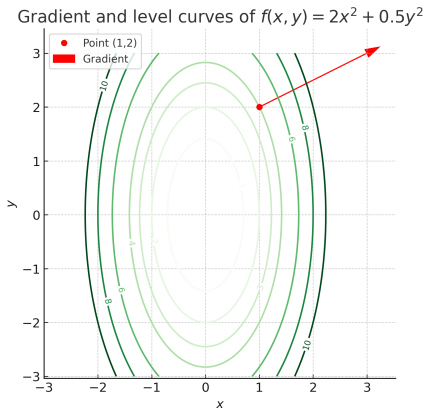
By Cauchy–Schwarz,

$$|D_{\mathbf{u}}f(\mathbf{x})| \leq \|\nabla f(\mathbf{x})\|,$$

with equality if $\mathbf{u}$ points in the same direction as $\nabla f(\mathbf{x})$.

Conclusion: $\nabla f(\mathbf{x})$ points in the direction of **steepest increase**, $-\nabla f(\mathbf{x})$ in the direction of **steepest decrease**.

Example: gradient and level curves



$$\text{Let } f(x, y) = 2x^2 + \frac{1}{2}y^2.$$

$$\nabla f(x, y) = (4x, y).$$

$$\text{At } (1, 2), \nabla f = (4, 2).$$

Geometry: The gradient is perpendicular to the level curve

$$2x^2 + \frac{1}{2}y^2 = c$$

through $(1, 2)$.

Note: ∇f is always normal to level sets. Why?

Jacobian and matrix differentiation rules

Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$. The **Jacobian** at $\mathbf{x}$ is

$$JF(\mathbf{x}) = \left[\frac{\partial F_i}{\partial x_j}(\mathbf{x}) \right]_{i=1, \dots, m; j=1, \dots, n} \in \mathbb{R}^{m \times n}.$$

If $m = 1$, $F = f$, then $Jf(\mathbf{x}) = \nabla f(\mathbf{x})^\top$.

Chain rule: If $G : \mathbb{R}^m \rightarrow \mathbb{R}^p$, $H = G \circ F$, then

$$JH(\mathbf{x}) = JG(F(\mathbf{x})) JF(\mathbf{x}).$$

Quick identities:

1. If $F(\mathbf{x}) = A\mathbf{x}$ with $A \in \mathbb{R}^{m \times n}$: $JF = A$.
2. If $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$: $\nabla f = \mathbf{a}$.
3. If $f(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x}$: $\nabla f = (A + A^\top)\mathbf{x}$ ($= 2A\mathbf{x}$ if A symmetric).

Unconstrained optimization

Local optima. $\mathbf{x}_0$ is a local max (min) if $f(\mathbf{x}) \leq (\geq) f(\mathbf{x}_0)$ nearby. If f is differentiable and $\mathbf{x}_0$ is an interior local extremum,

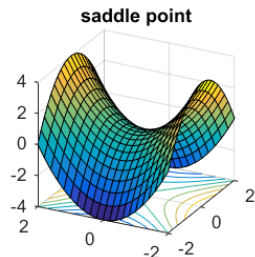
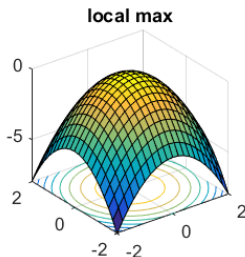
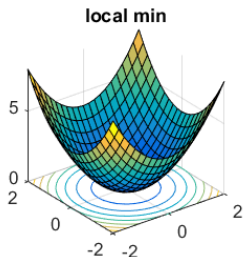
$$\nabla f(\mathbf{x}_0) = \mathbf{0} \quad (\text{first-order condition}).$$

If $f \in C^2$, Taylor expansion at a stationary point $\mathbf{x}_0$ gives

$$f(\mathbf{x}_0 + \mathbf{h}) = f(\mathbf{x}_0) + \frac{1}{2} \mathbf{h}^\top H_f(\mathbf{x}_0) \mathbf{h} + o(\|\mathbf{h}\|^2),$$

so the sign of the quadratic form $\mathbf{h}^\top H_f(\mathbf{x}_0) \mathbf{h}$ determines local behavior.

Unconstrained optimization (pictures)



Local optimality: second-order tests

Assume $f \in C^2$ and let $H_f(\mathbf{x}) = [f_{x_i x_j}(\mathbf{x})]_{i,j}$ be the (symmetric) Hessian.

At a stationary point $\mathbf{x}_0$:

- $H_f(\mathbf{x}_0)$ positive definite $\Rightarrow$ local minimum.
- $H_f(\mathbf{x}_0)$ negative definite $\Rightarrow$ local maximum.
- $H_f(\mathbf{x}_0)$ indefinite $\Rightarrow$ saddle.

$n = 2$ test: Let $D_2 = f_{xx}f_{yy} - f_{xy}^2$ at $\mathbf{x}_0$.

$D_2 > 0, f_{xx} > 0 \Rightarrow$ local min,

$D_2 > 0, f_{xx} < 0 \Rightarrow$ local max,

$D_2 < 0 \Rightarrow$ saddle, $D_2 = 0$: inconclusive.

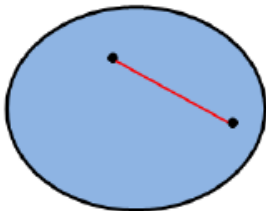
Examples

1. $f(x, y) = x^2 - y^2 - xy$. Then $\nabla f = (2x - y, -2y - x)$. The only stationary point is $(0, 0)$. The Hessian $H_f = \begin{pmatrix} 2 & -1 \\ -1 & -2 \end{pmatrix}$ is **indefinite** $\Rightarrow$ $(0, 0)$ is a **saddle**.
2. $f(x, y) = x^2 + y^4$. Stationary point: $(0, 0)$. $H_f(0, 0) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$ is positive semidefinite; $f \geq 0$ so $(0, 0)$ is a **global minimum**.
3. $f(x, y) = x^3 + y^3$. Stationary point: $(0, 0)$. The Hessian at $(0, 0)$ is 0; the point is a **saddle**.

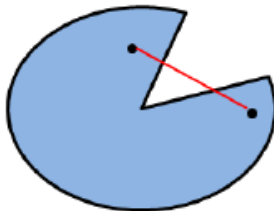
Convex domains

A set $D \subset \mathbb{R}^n$ is **convex** if for any $\mathbf{x}, \mathbf{y} \in D$ and $t \in [0, 1]$, the point $(1 - t)\mathbf{x} + t\mathbf{y} \in D$.

Convex



Non-convex



Convexity, concavity, and global optimality

Definition

On a convex domain $D \subset \mathbb{R}^n$, f is **convex** if

$$f(\theta \mathbf{x} + (1 - \theta) \mathbf{y}) \leq \theta f(\mathbf{x}) + (1 - \theta) f(\mathbf{y}), \quad \forall \mathbf{x}, \mathbf{y} \in D, \theta \in [0, 1].$$

Concave: reverse the inequality.

Curvature test (for C^2 functions)

$H_f(\mathbf{x}) \succeq 0 \quad \forall \mathbf{x} \in D \iff f$ convex on D , $H_f(\mathbf{x}) \preceq 0 \iff f$ concave.

Strict definiteness $\Rightarrow$ strict convexity/concavity.

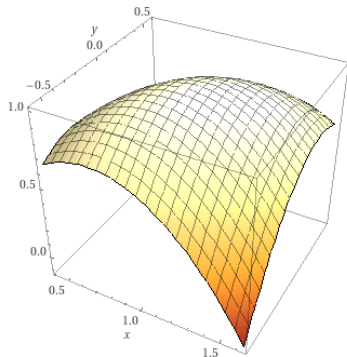
Composition rule: If g is convex and $f : \mathbb{R} \rightarrow \mathbb{R}$ is nondecreasing and convex, then $f \circ g$ is convex.

Example

Let $f(x, y) = 2x - y - x^2 + xy - y^2$. Then

$$\nabla f = (2 - 2x + y, -1 + x - 2y), \quad H_f = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}.$$

H_f is **negative definite** $\Rightarrow f$ is strictly concave. The unique stationary point solves $\nabla f = \mathbf{0}$, giving $(x, y) = (0, 1)$, which is a **global maximum**.



plot 2x-y-x^2+xy-y^2 | Computed by Wolfram|Alpha

Economic example: profit maximization

A firm sells products X/Y at 45/55 euros. Revenue $R(x, y) = 45x + 55y$.
Cost

$$C(x, y) = 300 + x^2 + 1.5y^2 - 25x - 35y.$$

Profit $f(x, y) = R(x, y) - C(x, y)$. Then

$$f_x = -2x + 70, \quad f_y = -3y + 90 \Rightarrow (x^*, y^*) = (35, 30).$$

Since $H_f = \begin{pmatrix} -2 & 0 \\ 0 & -3 \end{pmatrix}$ is negative definite everywhere, f is strictly concave and $(35, 30)$ is the **global maximum**. The maximal profit is $f(35, 30) = 2275$.

Least squares as orthogonal projection

Given data $X \in \mathbb{R}^{n \times d}$ and response $y \in \mathbb{R}^n$, the least-squares estimator solves

$$\hat{\beta} = \arg \min_{\beta} \|y - X\beta\|^2, \quad \hat{y} = X\hat{\beta}.$$

Geometric view (recall Lecture 2): $\hat{y}$ is the orthogonal projection of y onto the column space $\mathcal{C}(X)$, hence

$$X^\top(y - X\hat{\beta}) = 0 \iff (X^\top X)\hat{\beta} = X^\top y \quad (\text{if } X^\top X \text{ invertible}).$$

Polynomial regression: A common use of least squares is fitting nonlinear trends by expanding the design matrix X . For instance, with one predictor x , we can set

$$X = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^m \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^m \end{pmatrix},$$

so that the fitted model is $y \approx c_0 + c_1x + \cdots + c_mx^m$.

Ridge regression: stabilizing high-variance fits

When $X^\top X$ is ill-conditioned or d is large, add ℓ_2 regularization:

$$\hat{\beta}_\lambda = \arg \min_{\beta} (\|y - X\beta\|^2 + \lambda \|\beta\|^2) \implies \hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y.$$

Spectral view: if $X^\top X = U \text{diag}(s_1^2, \dots, s_d^2) U^\top$, then

$$\hat{\beta}_\lambda = \sum_{j=1}^d \frac{s_j}{s_j^2 + \lambda} u_j \langle y, Xu_j \rangle,$$

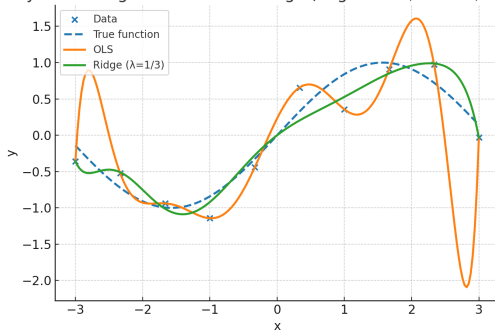
so ridge **shrinks** directions with small s_j (low variance) the most ($\frac{s_j}{s_j^2 + \lambda} < \frac{1}{s_j}$), reducing variance and overfitting.

Overfitting vs. ridge: degree-9 polynomial demo

$n = 10$ points from $y = \sin x + \varepsilon$ on $[-3, 3]$, degree 9 polynomial.

OLS (no penalty) vs. Ridge with $\lambda = \frac{1}{3}$.

Polynomial Regression: OLS vs Ridge (degree = 9, $n = 10$, $\sigma = 0.2$)



OLS wiggles violently between sparse points; ridge damps the high-frequency coefficients and tracks the signal more smoothly.

Modern applied examples (multivariable)

- Portfolio risk (mean–variance):

$$f(\mathbf{w}) = \mathbf{w}^\top \Sigma \mathbf{w}, \quad g(\mathbf{w}) = \mu^\top \mathbf{w}, \quad \mathbf{w} \in \mathbb{R}^n, \quad \sum_i w_i = 1, \quad w_i \geq 0.$$

- Logistic regression (binary choice):

$$J(\beta) = \frac{1}{n} \sum_{i=1}^n \left(\log(1 + e^{x_i^\top \beta}) - y_i x_i^\top \beta \right) \quad (\text{convex in } \beta).$$

- CES utility/production:

$$U(x) = \left(\sum_{i=1}^n \alpha_i x_i^\rho \right)^{1/\rho}, \quad P(L, K) = A(\theta) L^\alpha K^\beta.$$

Gradient descent

Goal: minimize $f(\theta)$.

$$\theta_{t+1} = \theta_t - \eta_t \nabla f(\theta_t).$$

Pieces you pick:

- **Step size** η_t : constant, diminishing, or via backtracking.
- **Stop** when $\|\nabla f(\theta_t)\|$ small.

In practice: feature scaling and a good η_t schedule matter a lot.

GD on least squares (closed form vs iterations)

Least squares problem has a closed form solution. This still requires inverting a potentially large matrix $X^\top X$. GD gives an alternative way to find a solution.

$$f(\beta) = \frac{1}{n} \|X\beta - \mathbf{y}\|^2, \quad \nabla f(\beta) = \frac{2}{n} X^\top (X\beta - \mathbf{y}).$$

GD update:

$$\beta_{t+1} = \beta_t - \eta \frac{2}{n} X^\top (X\beta_t - \mathbf{y}).$$

Ridge:

$$f_\lambda(\beta) = \frac{1}{n} \|X\beta - \mathbf{y}\|^2 + \lambda \|\beta\|_2^2, \quad \nabla f_\lambda = \frac{2}{n} X^\top (X\beta - \mathbf{y}) + 2\lambda\beta.$$

Closed form exists $(X^\top X)^{-1} X^\top \mathbf{y}$, but GD scales better to huge n, p or streaming data.

Constrained optimization: Lagrange and KKT (teaser)

Equality constraints $g_i(x) = 0$ and inequality constraints $h_j(x) \leq 0$. The Lagrangian:

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \mu_j h_j(x).$$

KKT conditions (when they apply):

- **Stationarity:** $\nabla_x \mathcal{L}(x^*, \lambda^*, \mu^*) = \mathbf{0}$.
- **Primal feasibility:** $g_i(x^*) = 0, h_j(x^*) \leq 0$.
- **Dual feasibility:** $\mu_j^* \geq 0$.
- **Complementary slackness:** $\mu_j^* h_j(x^*) = 0$.

Example (budgeted utility max): maximize $U(x)$ s.t. $p^\top x \leq B$. Then $\mathcal{L}(x, \mu) = U(x) + \mu(B - p^\top x)$ and at optimum $\nabla U(x^*) = \mu^* p, p^\top x^* \leq B, \mu^* \geq 0, \mu^*(B - p^\top x^*) = 0$.

When to use second order methods?

In general, we update θ_t as

$$\theta_{t+1} := \arg \min f(\theta_t) + \langle \nabla f(\theta_t), \theta \rangle + \frac{1}{2}(\theta - \theta_t)^\top \mathbf{K}(\theta - \theta_t).$$

If $\mathbf{K} = I_n$, we recover gradient descent.

If $\mathbf{K} = \nabla \nabla^\top f(\theta_t)$, we get the **Newton method**.

- **Newton:** $\theta_{t+1} = \theta_t - H^{-1}(\theta_t) \nabla J(\theta_t)$ (fast near solution, expensive to form/solve).
- **Quasi-Newton (e.g., BFGS, LBFGS):** approximate H^{-1} from gradients only; great for medium scale convex problems.
- **Takeaway:** for huge data/models use (S)GD; for smaller smooth convex problems, quasi-Newton shines.